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Abstract

AlphaZero has revolutionized AI game-playing, yet it may overlook actions crucial for long-term strategy. This paper introduces Entropy-Guided MCTS (EG-MCTS), enhancing AlphaZero by incorporating information theory into action selection. We propose an information gain metric based on game state entropy, integrating it into MCTS to balance immediate rewards with informational value. Our contributions include developing this metric, modifying neural network training, and systematically balancing information gain with existing criteria. Comprehensive evaluations across various games demonstrate EG-MCTS's improved performance in complex strategic scenarios, showing enhanced efficiency in game tree navigation and adaptation to unfamiliar situations. These findings suggest broader applications in sequential decision-making under uncertainty, advancing the development of more adaptable AI systems.

CCS Concepts

• Computing methodologies; **• Artificial intelligence**; **• Search methodologies**; **• Game tree search**; **• Machine learning**; **• Learning paradigms**; **• Reinforcement learning**;

Keywords

Monte Carlo Tree Search, Information Theory, Reinforcement Learning, Game AI

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1 Introduction

The field of game-playing algorithms has seen significant advancements, providing insights into decision-making and strategic planning [5]. Among these, the AlphaZero algorithm has demonstrated superhuman performance in chess, shogi, and Go by combining deep neural networks with Monte Carlo Tree Search (MCTS), showcasing the power of self-play reinforcement learning in mastering strategic domains [5, 6, 8, 11, 15].

Despite these accomplishments, optimizing the exploration-exploitation trade-off remains a challenge in reinforcement learning [9, 14, 20]. Current MCTS approaches balance estimated action values with visit frequency using upper confidence bound (UCB) formulas [2, 4, 10], but may miss actions that significantly alter the game state and provide long-term strategic advantages. This limitation is particularly critical in complex games where strategic depth is essential.

To address this, we propose Entropy-Guided MCTS (EG-MCTS), enhancing the AlphaZero framework by incorporating information gain principles from information theory into the MCTS selection process [3]. By quantifying the uncertainty or information content of game states, EG-MCTS adds a new dimension to action selection, beyond immediate value estimates and exploration bonuses. As illustrated in Figure 1, this approach aims to enhance decision-making by navigating complex game trees more efficiently, potentially reducing search depth, identifying key moves that significantly impact the game's trajectory even if their immediate value is not apparent, and adapting quickly to unfamiliar or strategic situations by prioritizing high-information actions.

Our contributions include:

- Balancing information gain with existing value and exploration terms in action selection of MCTS.
- Modifying neural network training to account for informational value.
- Providing empirical evaluations on Gomoku game environments, demonstrating EG-MCTS's effectiveness.

The implications extend beyond game-playing AI to fields like robotics and autonomous systems, where sequential decision-making under uncertainty is crucial. By integrating information-theoretic principles into reinforcement learning, we aim to develop AI agents that not only play to win but also to understand, unlocking new levels of strategic depth and adaptability.

In this paper, we present the related work on MCTS, entropy based exploration and alphazero and its variants (Section 2), the background on AlphaZero, MCTS, and information theory (Section 3), the implementation of EG-MCTS (Section 4), a comprehensive empirical evaluation (Section 5), and conclude with a summary of contributions and their significance (Section 6). This work aims to advance game-playing AI and the development of sophisticated decision-making algorithms, bridging information theory and reinforcement learning for more adaptable AI agents.

2 Related work

This section reviews related work in the areas of MCTS, entropy-based exploration methods, and recent advancements in AlphaZero-like algorithms.

2.1 Monte Carlo Tree Search (MCTS)

MCTS has been a pivotal algorithm in game AI, providing a robust framework for decision making in uncertain environments [17]. The seminal work by Kocsis and Szepesvári [4] introduced the Upper Confidence Bounds for Trees (UCT) method, which balances exploration and exploitation during search. Coulom [2] demonstrated the effectiveness of MCTS in the game of Go, leading to significant improvements over traditional heuristic-based methods. Recent enhancements to MCTS, such as PUCT (Predictor + UCT) [26] used in AlphaGo [5], have further refined the algorithm by incorporating policy priors from neural networks. [1] provide a comprehensive review of recent modifications and applications of MCTS, highlighting its versatility and continued relevance in modern AI research.

2.2 Entropy-Based Exploration

Entropy-based exploration strategies have been explored in various reinforcement learning contexts [18]. Entropy, a measure of uncertainty or information content, has been used to encourage exploration in environments where uncertainty about the outcome of actions is high. For instance, Houthooft et al. [3] proposed VIME, which uses information gain to guide exploration in reinforcement learning tasks. This approach rewards the agent for visiting states that maximize the reduction in uncertainty about the environment's dynamics. Similarly, Houlsby et al. [21] and Hernández-Lobato et al. [22] have applied information-theoretic principles to guide exploration in active learning and Bayesian optimization, respectively. These approaches demonstrate the potential of entropy-based methods to improve exploration efficiency and decision-making in complex environments.

2.3 AlphaZero and Its Variants

AlphaZero, developed by Silver et al. [5], is a generalized version of AlphaGo Zero, capable of mastering multiple games using self-play and reinforcement learning. The architecture of AlphaZero integrates a dual-headed neural network with MCTS, where the neural network provides both policy and value predictions. Recent research has focused on enhancing AlphaZero's performance through various modifications. MuZero [6] extends AlphaZero by learning a model of the environment's dynamics, allowing it to handle environments with unknown dynamics. Schrittwieser et al. [6] demonstrated the effectiveness of this approach in mastering Atari, Go, chess, and shogi. Additionally, Niu et al. [7] introduced LightZero, a benchmark for MCTS in general sequential decision scenarios, further expanding the applicability of AlphaZero-like algorithms.

Our work builds on these foundational advancements by integrating an entropy-based information gain metric into the MCTS framework of AlphaZero. This novel approach aims to improve the efficiency of the exploration process by prioritizing actions that lead to significant changes in the game state, thereby reducing uncertainty and enhancing long-term strategic decision making.

3 Background

This section provides the theoretical foundation for our proposed Entropy-Guided Monte Carlo Tree Search (EG-MCTS) algorithm. We begin by reviewing the key components of the AlphaZero framework, followed by a detailed exploration of Monte Carlo Tree Search (MCTS) and its application in game-playing AI. We then introduce relevant concepts from information theory, particularly focusing on entropy and its role in decision-making processes. Finally, we discuss the intersection of these areas, setting the stage for our novel approach.

3.1 AlphaZero Framework

AlphaZero represents a significant advancement in artificial intelligence, combining deep neural networks with MCTS to master complex perfect information games [5]. At its core, AlphaZero utilizes a dual-headed neural network $f_\theta(s)$ that, given a game state s, outputs: A) A policy vector $p = (p_1, \ldots, p_n)$, where p_i represents the probability of selecting action a_i in state s. B) A scalar value $v \in [-1, 1]$, estimating the expected outcome of the game from state s.

Formally, we can express this as:

$$(p, v) = f_\theta(s) \tag{1}$$

where θ represents the parameters of the neural network.

The network is trained through self-play reinforcement learning, iteratively improving its ability to evaluate game states and select promising actions.

3.2 Monte Carlo Tree Search (MCTS)

MCTS is a heuristic search algorithm that has shown remarkable effectiveness in decision-making processes, particularly in game-playing AI [1, 12]. The algorithm builds and updates a search tree iteratively, with each node representing a game state and edges representing actions.

In the context of AlphaZero, MCTS is used to guide the exploration of the game tree. The process consists of four main steps:

Selection: Traverse the tree from the root to a leaf node using a selection policy.

Expansion: Add one or more child nodes to the selected leaf node.

Simulation: Perform a rollout from the new node(s) to estimate the value of the state.

Backpropagation: Update the statistics of the nodes in the path from the new node to the root.

The selection policy in AlphaZero uses an Upper Confidence Bound (UCB) formula [4, 10, 16]:

$$a^* = argmax\,[Q(s,a) + U(s,a)] \tag{2}$$

where $Q(s,a)$ is the estimated action value, and U(s,a) is an exploration term defined as:

$$U(s,a) = c_{puct} * P(s,a) * \left(\frac{\sqrt{\sum_b N(s,b)}}{1 + N(s,a)}\right) \tag{3}$$

Here, c_{puct} is a constant that controls the level of exploration, $P(s,a)$ is the prior probability of action a in state s (obtained from the policy network), and $N(s,a)$ is the visit count for state-action pair (s,a).

3.3 Information Theory in Decision Making

Information theory, introduced by Claude Shannon [23], provides a mathematical framework for quantifying information. In the context of decision-making and game-playing AI, two key concepts are particularly relevant:

- 1. Entropy: A measure of the average information content or uncertainty in a probability distribution. For a discrete probability distribution $P(X)$, the entropy $H(X)$ is defined as:

$$H(X) = -\sum_x P(x) \cdot logP(x) \tag{4}$$

- 2. Information Gain: The reduction in entropy (or uncertainty) after obtaining new information. Given a variable Y, the information gain $IG(X|Y)$ is the difference in entropy before and after observing Y:

$$IG(X|Y) = H(X) - H(X|Y) \tag{5}$$

In the context of game-playing, we can interpret the entropy of a game state as a measure of its strategic complexity or the uncertainty in the outcome [3]. States with high entropy may represent critical junctures in the game where multiple viable options exist, while low entropy states may have clearer optimal actions.

4 Connecting MCTS and Information Theory

The integration of information-theoretic concepts into MCTS presents an opportunity to enhance the algorithm's decision-making process. By considering the information gain of actions in addition to their estimated value and exploration bonus, we can potentially guide the search towards more informative and strategically significant moves [13, 19].

This approach aligns with recent work in active learning and Bayesian optimization, where information-theoretic acquisition functions have been used to guide sampling strategies [21, 22]. In the context of game-playing AI, similar principles can be applied to action selection in MCTS. The challenge lies in effectively quantifying the information gain of actions in the game tree and incorporating this metric into the existing MCTS framework. Our proposed Entropy-Guided MCTS algorithm, detailed in the next section, addresses this challenge by introducing an entropy-based information gain term into the action selection criteria.

By leveraging these foundational concepts from AlphaZero, MCTS, and information theory, we aim to develop a more nuanced and adaptable approach to game tree exploration, potentially leading to improved performance in complex strategic scenarios.

5 Entropy-Guided Monte Carlo Tree Search (EG-MCTS)

In this section, we present our novel Entropy-Guided Monte Carlo Tree Search (EG-MCTS) algorithm, which extends the traditional AlphaZero framework by incorporating an information gain metric into the MCTS selection process. The overall framework is illustrated in Figure 1.

5.1 Algorithm Overview

EG-MCTS builds upon the AlphaZero framework by introducing an entropy-based information gain term into the action selection criteria of MCTS. The core idea is to balance the exploitation of high-value actions, exploration of less-visited nodes, and the potential information gain of actions that could significantly alter the game state. Algorithm 1 provides a high-level overview of EG-MCTS.

Algorithm 1 Entropy-Guided Monte Carlo Tree Search (EG-MCTS)

$function\ EG - MCTS(s_0,\ f_\theta,\ N)$
  $Initialize\ search\ tree\ T\ with\ root\ node\ s_0$
  $for\ i = 1\ to\ N\ do$
    $s \leftarrow s_0$
    $while\ s\ is\ not\ a\ leaf\ node\ do$
      $a \leftarrow SelectAction(s)$
      $s \leftarrow ChildState(s,\ a)$
    $(p,\ v) \leftarrow f_\theta(s)$
    $ExpandNode(s,\ p)$
    $\Delta \leftarrow v$
    $BackpropagateValue(s,\ \Delta)$
  $return\ NormalizedVisitCount(s_0)$
$function\ SelectAction(s)$
  $return\ argmax_a\ [Q(s,a) + U(s,a) + \lambda \cdot IG(s,a)]$
$function\ IG(s,\ a)$
  $s' \leftarrow ChildState(s,\ a)$
  $(p,v) \leftarrow f_\theta(s)$
  $(p',\ v') \leftarrow f_\theta(s')$
  $return\ (1 - v') \cdot log(1 - v') - v \cdot log(v)$

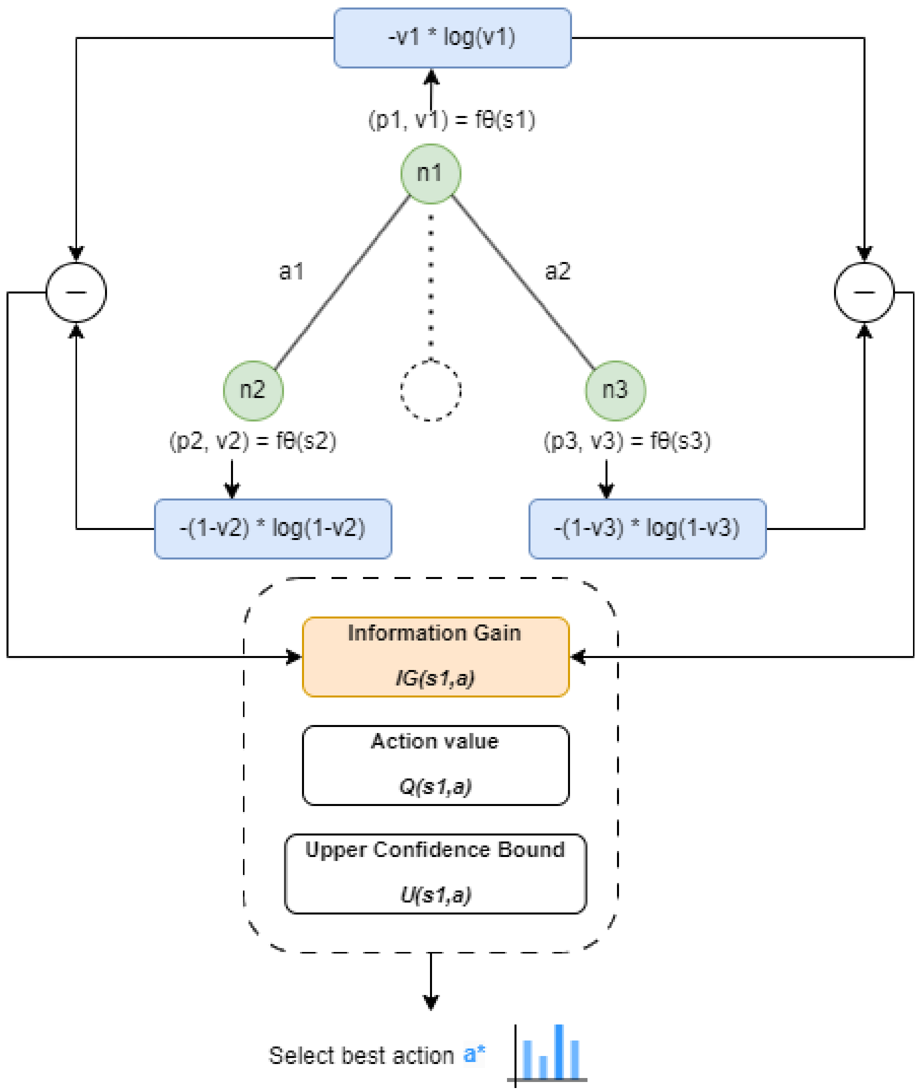

Figure 1: Enhancing MCTS Selection Process with Information Gain. Starting from root node n_1 with state s_1, moving probability p_1, and winning probability v_1 for player 1. The entropy at n_1 is calculated as $-v_1 log(v_1)$. Possible subsequent nodes n_2 and n_3 are shown. For node n_2, the winning probability for player 2 is v_2, and for player 1, it is $1 - v_2$, allowing the calculation of entropy and information gain. The information gain metric $IG(s_1,\ a)$ is integrated with the action value $Q(s_1,\ a)$ and upper confidence bound $U(s_1,\ a)$ to select the best action a^*.

The key modification in EG-MCTS compared to standard MCTS is the inclusion of the information gain term $IG(s, a)$ in the action selection criteria (line 14).

5.2 Information Gain Calculation

The information gain of an action a in state s is defined as the difference in entropy between the current state s and the resulting state $s^{'}$ after taking action a:

$$IG\,(s, a) \;=\; H\,(s) \;-\; H\left(s^{'}\right) \tag{6}$$

where H(s) represents the entropy of a state. To calculate this entropy, we leverage the value prediction v from the neural network f_θ, interpreting it as a proxy for the state's winning probability:

$$H\,(s) \;=\; -v \;\cdot\; log\,(v) \;-\; (1-v) \;\cdot\; log\,(1-v) \tag{7}$$

Substituting these into equation (1), we arrive at our final information gain formula:

$$IG\,(s, a) = \left(1 - v^{'}\right) \cdot log\left(1 - v^{'}\right) - v \cdot log\,(v) \tag{8}$$

where v and v' are the value predictions for states s and s', respectively.

5.3 Enhanced MCTS Selection

We modify the MCTS selection criteria to incorporate the information gain:

$$a^* = argmax_a \ [Q(s,a) + U(s,a) + \lambda \cdot IG(s,a)] \tag{9}$$

Here, λ is a hyperparameter that controls the weight of the information gain term. The $Q(s,a)$ and $U(s,a)$ terms remain as defined in the original AlphaZero algorithm.

5.4 Neural Network Adaptation

To fully leverage the information gain concept, we propose augmenting the neural network training process. We introduce an additional loss term LIG that encourages the network to accurately predict states with high information gain potential:

$$LIG = -\sum_{s,a} p(a|s) \cdot IG(s,a) \tag{10}$$

where $p(a|s)$ is the policy output of the network for action a in state s. The total loss function then becomes:

$$L = L_v + L_p + \alpha \cdot LIG \tag{11}$$

where L_v and L_p are the original value and policy losses, respectively, and α is a hyperparameter controlling the influence of the information gain loss.

This enhanced algorithm aims to guide the search towards informative actions, potentially improving performance in complex scenarios that require long-term planning and positional understanding.

6 Experiments

6.1 Experimental Setting

We applied the EG-MCTS algorithm to the game of Gomoku (also known as Five in a Row) as a benchmark for challenging planning problems that require strategic thinking and long-term planning. Gomoku was chosen for its balance of simplicity in rules and complexity in strategy, making it an ideal testbed for our information-theoretic approach.

Following the methodology of MuZero [6], we trained our EG-MCTS model for 1 million mini-batches of size 2048. During both training and evaluation, EG-MCTS used 800 simulations for each search, mirroring the setup used in board games by MuZero.The neural network architecture in EG-MCTS consists of: A) a representation function using the same convolutional [24] and residual [25] architecture as AlphaZero, but with 16 residual blocks. B) dynamics function using the same architecture as the representation function. C) a prediction function using the same architecture as AlphaZero.All networks use 256 hidden planes. The key difference in our setup is the introduction of the information gain term in the MCTS selection process and the additional loss term in the neural network training, as described in Section 3.

We compared EG-MCTS against three baselines: A) standard AlphaZero algorithm; B) a variant of AlphaZero with increased exploration (higher cpuct); C) a random policy baseline.

Table 1: Win rates of EG-MCTS against baselines (%)

Opponent	Win	Draw	Loss
Standard AlphaZero	62	15	23
Increased Explore	58	18	24
Random Policy	99	1	0

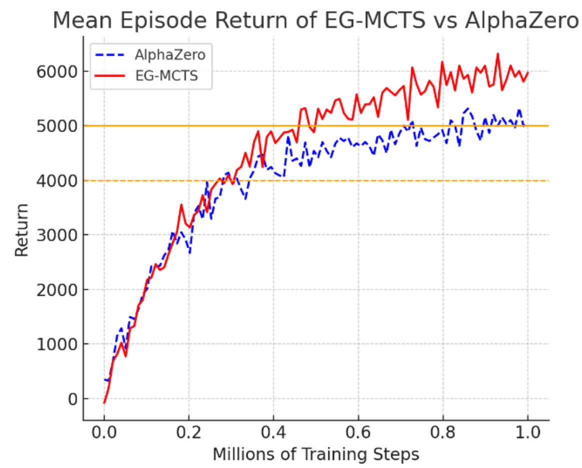

Figure 2: Mean episode return of EG-MCTS vs AlphaZero

7 Results

7.1 Overall Performance.

EG-MCTS demonstrated superior performance compared to all baselines, achieving a win rate of 62% against standard AlphaZero and 58% against the increased exploration variant. Table 1 summarizes these results.

7.2 Learning Efficiency.

EG-MCTS showed faster learning, reaching the same level of play as AlphaZero in fewer training steps. Figure 2 illustrates the mean episode return of EG-MCTS and AlphaZero.

8 Conclusion

In this paper, we introduced Entropy-Guided Monte Carlo Tree Search (EG-MCTS), a novel algorithm that enhances the MCTS process with information-theoretic principles. By incorporating an information gain term into the action selection criteria, EG-MCTS demonstrates improved performance in the game of Gomoku, particularly in complex strategic scenarios.Our experimental results show that EG-MCTS outperforms standard AlphaZero in terms of overall win rate, and learning efficiency.

The success of EG-MCTS in Gomoku suggests that information-theoretic approaches can offer significant benefits in game-playing AI and potentially in broader sequential decision-making problems. Future work could explore the application of these principles to a wider range of games and real-world decision-making scenarios.

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